■ TrialLink

RAG Pipeline — Evaluation Report

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MLOps Course IE-7374 | Northeastern University | Khoury College of Computer Sciences

1. Executive Summary

TrialLink is a clinical trial matching platform that uses a Retrieval-Augmented Generation (RAG) pipeline to match patients to relevant clinical trials. This report evaluates the performance of the RAG pipeline on **45 synthetic patients** generated using Synthea, covering breast cancer and diabetes conditions. The pipeline achieved an **eligible rate of 80.0%** and a **borderline rate of 4.4%**, with a guardrail flag rate of 20.0%.

2. Key Performance Metrics

Total Patients	Eligible	Borderline
45	36	2
Ineligible	**Blocked**	**Eligible Rate**
0	7	80.0%
Flag Rate	**Avg Trials Retrieved**	**Consensus Rate**
20.0%	4.9	0.0%
Input LLM Errors	**Output LLM Errors**	**Policy Flags**
1	7	0

3. Dataset

Synthetic patients were generated using the Synthea patient generator and parsed using the TrialLink FHIRParser. Patients are categorized by their primary condition:

Condition	Patients	Eligible	Borderline	Ineligible	Blocked	Eligible Rate	Avg Retrieved
Breast Cancer	9	6	1	0	2	67%	5.0

Condition	Patients	Eligible	Borderline	Ineligible	Blocked	Eligible Rate	Avg Retrieved
Diabetes	36	30	1	0	5	83%	4.9

Note: 'Other' condition patients should receive zero trial suggestions — TrialLink only supports breast cancer and diabetes.

4. Verdict Distribution

Each patient is assigned an overall verdict: ELIGIBLE (at least one trial fully matched), BORDERLINE (partial match requiring clinician review), or BLOCKED (guardrail intervention).

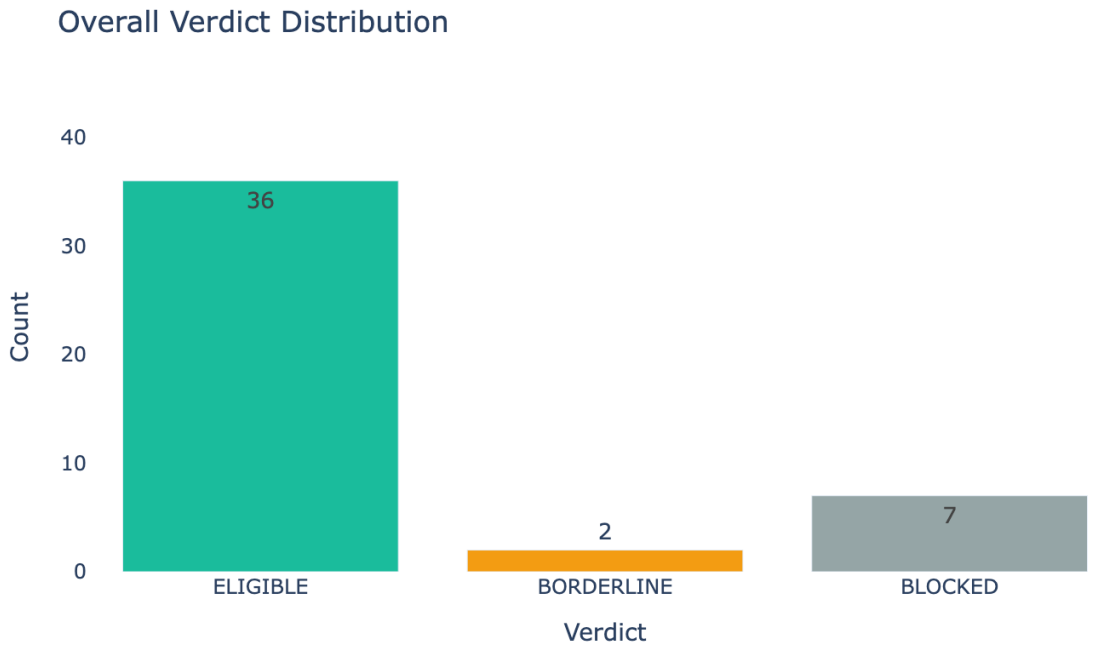

Figure 1: Overall verdict distribution across all patients.

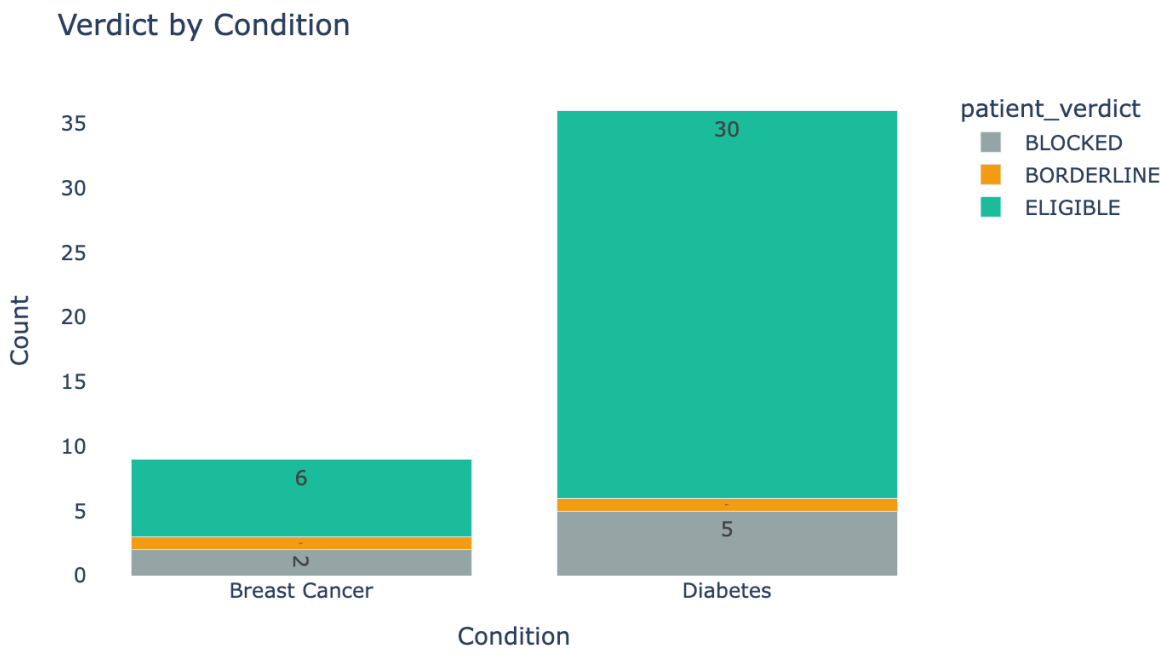

Figure 2: Verdict distribution broken down by patient condition.

5. Condition Segregation Validation

TrialLink enforces that patients with unsupported conditions receive no trial suggestions. The validate_trials_scope() guardrail and condition-aware Firestore collection filtering enforce this at both retrieval and output stages.

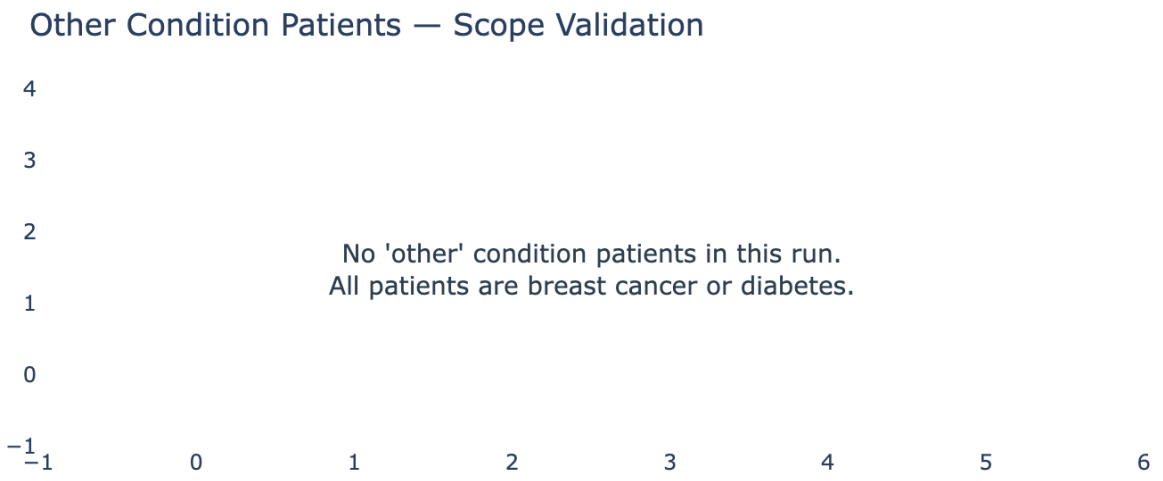

Figure 3: Scope validation for non-supported condition patients.

6. Guardrail Analysis

The pipeline implements a multi-layer guardrail system: PII redaction, structural validation, LLM input/output judges, policy checks, and grounding verification.

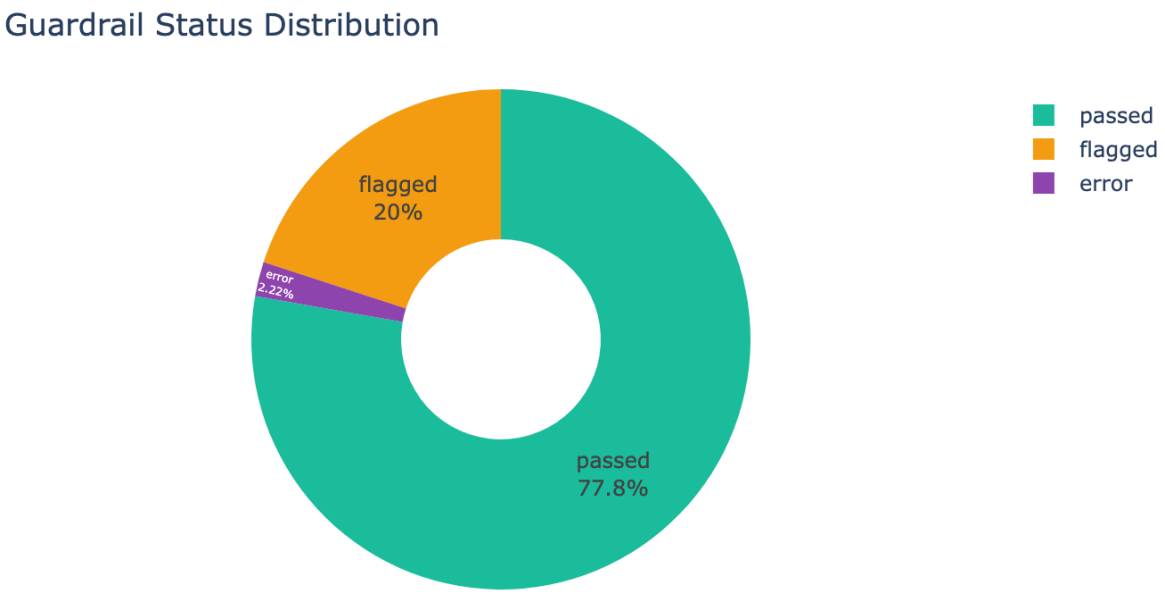

Figure 4: Guardrail status distribution (passed / flagged / blocked).

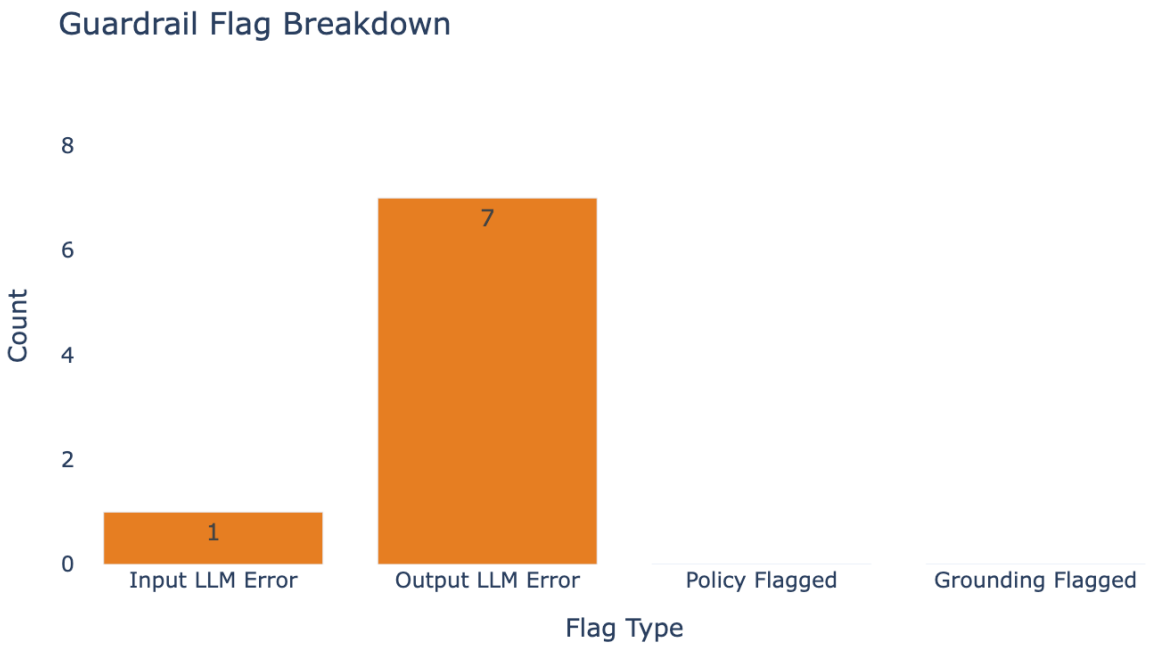

Figure 5: Guardrail flag breakdown by type.

- **Input LLM Guardrail Errors:** 1 — caused by JSON truncation in guardrail response.
- **Output LLM Guardrail Errors:** 7 — same root cause.
- **Policy Flags:** 0 — triggered by medication name patterns.

7. Retrieval Quality

The hybrid retrieval pipeline uses LLM-based clinical context enrichment to build targeted search queries, Vertex AI Vector Search (text-embedding-005) for dense retrieval, condition-aware Firestore collection filtering using the engineered 'disease' field, and Vertex AI Ranking API for reranking. This eliminates cross-condition contamination (breast cancer trials appearing for diabetes patients).

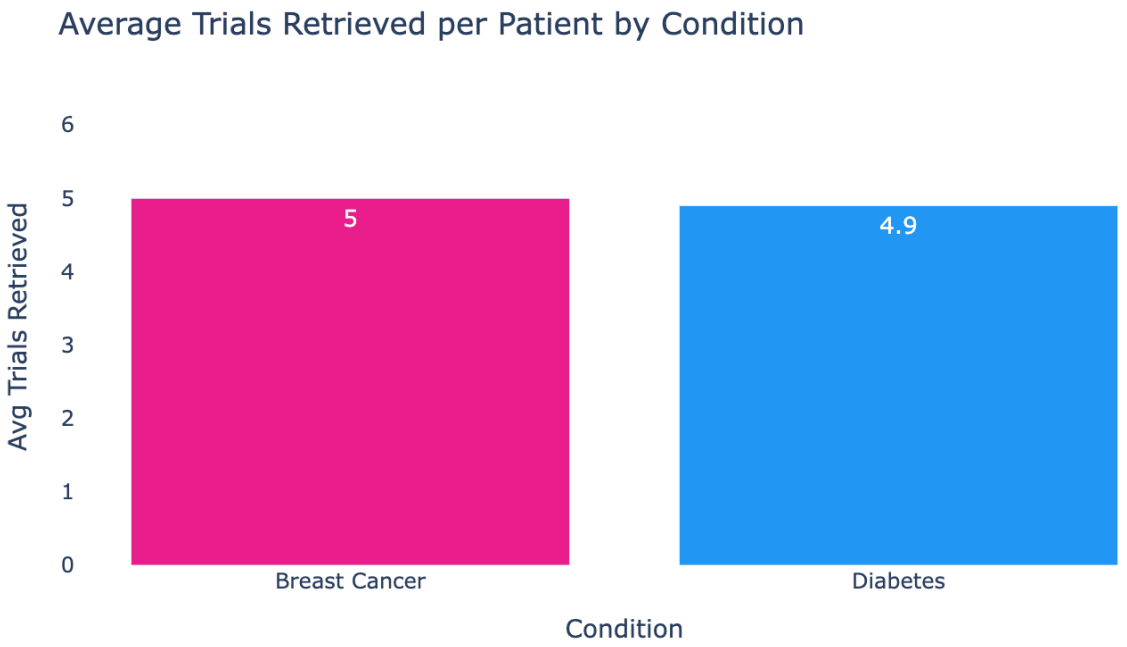

Figure 6: Average number of trials retrieved per patient by condition.

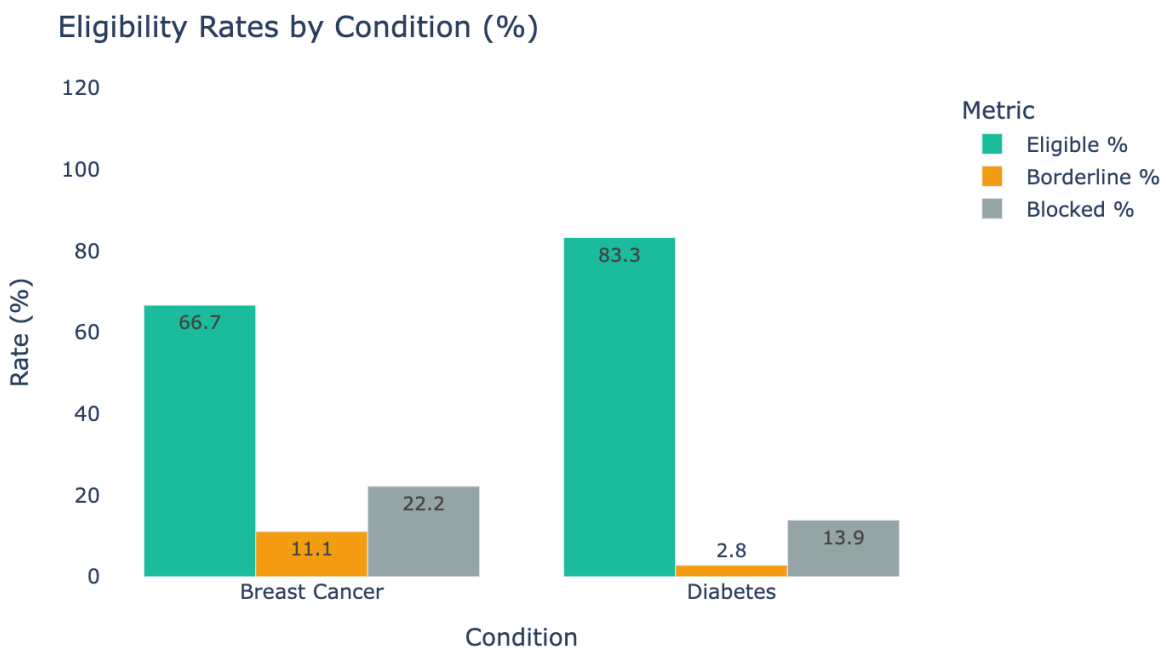

Figure 7: Eligibility rates (%) broken down by condition.

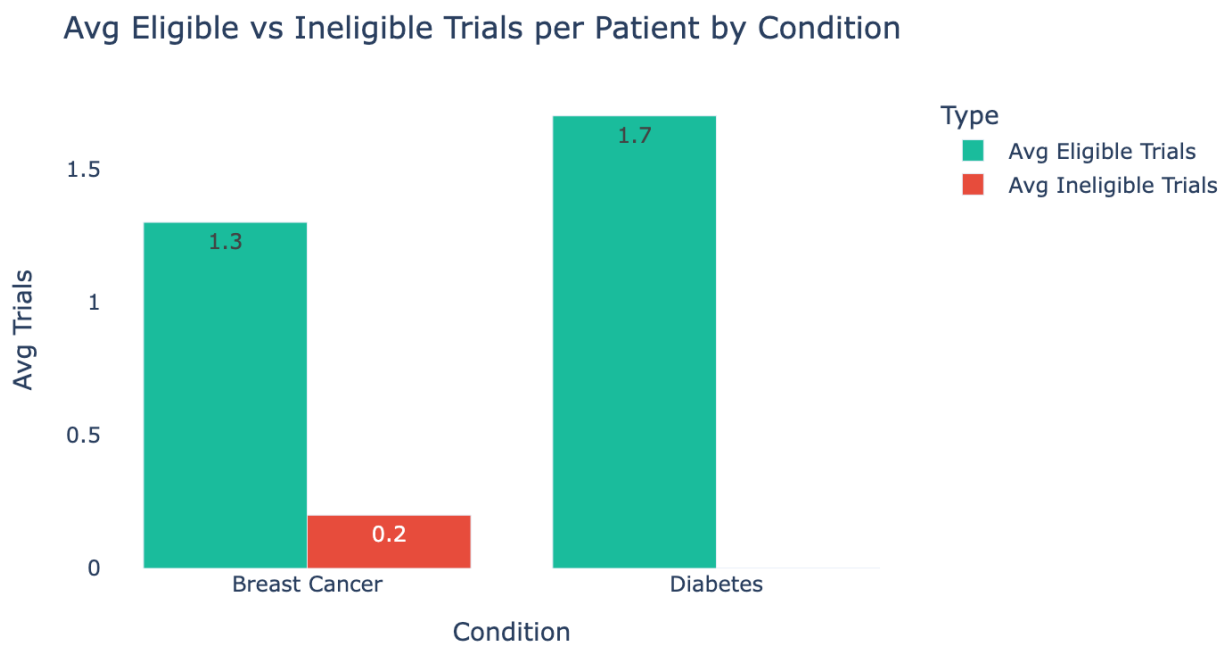

Figure 8: Average eligible vs ineligible trials per patient by condition.

8. Dual LLM Judge — Gemini vs MedGemma

TrialLink uses a dual-judge architecture. Gemini 2.5 Flash generates the primary eligibility recommendation. MedGemma (a medical-domain specialist LLM deployed on Vertex AI) acts as an independent second-opinion judge. A trial is only shown to the patient if BOTH judges agree it is ELIGIBLE or BORDERLINE.

- MedGemma was available for **0** of 45 patients in this run.
- Gemini-MedGemma consensus rate: **0.0%**
- MedGemma dedicated endpoints are only accessible within the GCP VPC. Local evaluation uses Gemini-only fallback. MedGemma is validated separately via Cloud Shell.

9. Per-Patient Results

Patient ID	Condition	Verdict	Guardrail	Retrieved	Eligible	Borderline
04d2dd7a	diabetes	ELIGIBLE	passed	5	4	0
04f3e444	diabetes	BLOCKED	passed	5	0	0
15167c18	diabetes	ELIGIBLE	passed	5	3	1
18bda507	diabetes	ELIGIBLE	passed	5	1	0
1d70947f	diabetes	ELIGIBLE	passed	5	4	0
23704fc4	diabetes	ELIGIBLE	passed	5	3	0
30a1b6de	diabetes	ELIGIBLE	passed	5	2	0

Patient ID	Condition	Verdict	Guardrail	Retrieved	Eligible	Borderline
32d11e41	diabetes	ELIGIBLE	passed	5	3	0
3f6e11b4	breast_cancer	ELIGIBLE	flagged	5	3	0
4a037d2b	diabetes	BORDERLINE	passed	5	0	1
4c8022e6	diabetes	ELIGIBLE	passed	5	1	0
4e298800	diabetes	BLOCKED	passed	5	0	0
5bfdbb40	diabetes	ELIGIBLE	passed	5	5	0
5f9e602f	diabetes	ELIGIBLE	passed	5	1	0
6e0f8854	diabetes	ELIGIBLE	passed	5	1	0
6e3bb83f	breast_cancer	ELIGIBLE	passed	5	3	0
7142b2d5	diabetes	ELIGIBLE	passed	5	2	0
724c5db9	diabetes	ELIGIBLE	passed	5	1	0
776898dd	diabetes	ELIGIBLE	passed	5	1	0
7e65f42a	diabetes	ELIGIBLE	flagged	5	2	0
824a9ab9	breast_cancer	ELIGIBLE	passed	5	2	0
82fba2b7	diabetes	ELIGIBLE	passed	5	3	1
85309a5c	diabetes	ELIGIBLE	flagged	5	2	0
8dca30f7	diabetes	ELIGIBLE	passed	5	1	0
93c189da	diabetes	ELIGIBLE	passed	5	1	0
94a45350	diabetes	ELIGIBLE	passed	5	2	0
96f10015	diabetes	BLOCKED	error	0	0	0
97be3445	diabetes	ELIGIBLE	flagged	5	4	0
97bed22e	diabetes	BLOCKED	passed	5	0	0
9ec69569	breast_cancer	BORDERLINE	passed	5	0	2
ad758292	diabetes	ELIGIBLE	passed	5	1	0
ae4ffadf	diabetes	ELIGIBLE	passed	5	1	0
b17d2697	diabetes	BLOCKED	flagged	5	0	0
b82e052e	breast_cancer	BLOCKED	passed	5	0	0
b889f1a4	diabetes	ELIGIBLE	passed	5	1	0
c98be91c	diabetes	ELIGIBLE	passed	5	2	0

Patient ID	Condition	Verdict	Guardrail	Retrieved	Eligible	Borderline
ceec0dea	diabetes	ELIGIBLE	flagged	5	2	0
d1693b81	breast_cancer	ELIGIBLE	passed	5	1	1
d2b3a72a	breast_cancer	BLOCKED	passed	5	0	0
d7119618	diabetes	ELIGIBLE	passed	5	1	0
d8e576a8	diabetes	ELIGIBLE	flagged	5	1	0
e0c351d9	diabetes	ELIGIBLE	passed	5	2	0
eb019545	diabetes	ELIGIBLE	flagged	5	2	0
eeaaf204	breast_cancer	ELIGIBLE	flagged	5	2	0
fb9c70bb	breast_cancer	ELIGIBLE	passed	5	1	0